



Statistical Distributions · Worksheet 1

Statistics Chapter 6: Statistical Distributions. Answer all questions. Show your working clearly. Total: 103 marks. Time: 2 hours 10 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are five sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Always **define** your random variable and state its distribution, e.g. $X \sim B(20, 0.3)$. Give answers to 4 decimal places or 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Discrete random variables	20	
B The binomial distribution	22	
C The normal distribution	19	
D The inverse normal and finding μ and σ	20	
E Choosing and evaluating a model	18	
Roar-level challenge	4	
Total	103	

Section A Discrete random variables

20 marks

1. The discrete random variable X has the probability distribution below.

[3]

x	1	2	3	4
$P(X = x)$	0.2	k	0.3	0.2

Find (a) the value of k , (b) $P(X \geq 2)$, (c) $P(1 < X < 4)$.

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2. The random variable X has probability function $P(X = x) = kx$ for $x = 1, 2, 3, 4$. [3]
Find the value of k and $P(X > 2)$.

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3. The random variable X takes the values 0, 1, 2 and 3 only, with $P(X = 0) = 0.1$, $P(X = 1) = a$, [3]
 $P(X = 2) = b$ and $P(X = 3) = 0.2$.
Given that $P(X \leq 1) = 0.45$, find a and b .

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4. Two fair four-sided spinners, each numbered 1 to 4, are spun. The random variable X is the larger of [4]
the two scores (or the common score if they are equal).
(a) Find the probability distribution of X .
(b) Find $P(X \geq 3)$.

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5. The discrete random variable X has probability function

$$P(X = x) = \begin{cases} k(5 - x) & x = 1, 2, 3, 4 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k = 0.1$. [2]

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(b) Write the probability distribution of X in a table. [1]

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(c) Find $P(X \leq 2)$. [1]

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(d) Two independent observations X_1 and X_2 are made. Find $P(X_1 + X_2 = 5)$. [3]

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(Total for Question 5 is 7 marks)

**Section B The binomial distribution**

22 marks

6. $X \sim B(10, 0.3)$. Find $P(X = 3)$.

[2]

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7. $X \sim B(15, 0.4)$. Find (a) $P(X = 4)$, (b) $P(X \leq 4)$, (c) $P(X > 7)$.

[3]

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8. 20% of the chocolates made by a company contain a nut. A bag holds 50 chocolates. X is the number of chocolates in the bag that contain a nut.

[3]

(a) Write down the mean number of chocolates in the bag that contain a nut.

(b) Find $P(X = 10)$ and $P(X \leq 8)$.

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9. $X \sim B(20, 0.35)$. Find (a) $P(X \leq 5)$, (b) $P(X \geq 8)$, (c) $P(3 < X \leq 8)$.

[4]

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10. A gardener plants seeds in trays of 20. Each seed germinates with probability 0.85, independently of the other seeds.

- (a) State two conditions needed for the number of seeds that germinate in a tray to be modelled by a binomial distribution, in context. [2]

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- (b) Find the probability that exactly 18 seeds in a tray germinate. [2]

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- (c) Find the probability that at most 15 seeds in a tray germinate. [2]

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- (d) Find the probability that at least 17 seeds in a tray germinate. [2]

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- (e) The gardener plants 6 trays. Find the probability that at least 5 of the trays have at least 17 seeds that germinate. [2]

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(Total for Question 10 is 10 marks)



Section C The normal distribution

19 marks

11. $X \sim N(50, 4^2)$. Find $P(X < 54)$. [2]

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12. $X \sim N(20, 3^2)$. Find (a) $P(X > 24)$, (b) $P(17 < X < 23)$. [3]

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13. IQ scores are modelled by $X \sim N(100, 15^2)$. Find (a) $P(85 < X < 130)$, (b) $P(X > 130)$. [3]

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14. The times taken to run a race are normally distributed with mean 30 minutes and standard deviation 2 minutes. [3]

(a) Without a calculator, write down an interval that contains approximately 95% of the times.

(b) Write down $P(X = 30)$, giving a reason.

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15. The masses of apples from an orchard are modelled as $X \sim N(150, 12^2)$, in grams. Apples heavier than 170 g are graded “large”.

(a) Find the proportion of apples that are graded large. [2]

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(b) Find $P(140 < X < 165)$. [2]

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(c) A box contains 10 apples chosen at random. Find the probability that at least one of them is large. [2]

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(d) The lightest 10% of apples are used for juice. Find the greatest mass of an apple used for juice. [2]

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(Total for Question 15 is 8 marks)



Section D The inverse normal and finding μ and σ

20 marks

16. $Z \sim N(0, 1)$. Find the value of z such that $P(Z < z) = 0.9$.

[2]

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17. $X \sim N(60, 8^2)$. Find the value of a such that $P(X < a) = 0.95$.

[3]

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18. $X \sim N(\mu, 5^2)$ and $P(X > 55) = 0.1$. Find μ .

[3]

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19. $X \sim N(30, \sigma^2)$ and $P(X < 40) = 0.945$. Find σ .

[4]

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20. The times, X minutes, that customers spend in a shop are modelled by a normal distribution. 10% of customers spend less than 35 minutes and 5% spend more than 70 minutes.

(a) Find the mean and standard deviation of X .

[5]

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(b) Find the proportion of customers who spend between 40 and 55 minutes in the shop.

[3]

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(Total for Question 20 is 8 marks)



Section E Choosing and evaluating a model

18 marks

21. State whether a binomial or a normal distribution is more suitable to model (a) the number of sixes in 20 rolls of a die, (b) the heights of adult men. [2]

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22. Five cards are taken without replacement from a pack of 52. Explain why the number of aces taken cannot be modelled by $B(5, \frac{1}{13})$. [2]

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23. The lengths of 200 bolts have mean 25.1 mm, median 25.0 mm, standard deviation 0.4 mm, minimum 23.9 mm and maximum 26.2 mm. [3]
Give three reasons why a normal distribution may be a suitable model for the lengths.

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24. The times, X minutes, that patients wait at a clinic are modelled by $X \sim N(5, 4^2)$. [3]
Find $P(X < 0)$ and use your answer to comment on the suitability of the model.

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25. A student models the daily total rainfall, R mm, at a location in the large data set by $R \sim N(1.8, 3.2^2)$.

(a) Find $P(R < 0)$, and explain why this shows the model is not suitable.

[3]

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(b) Suggest, with a reason, what the shape of the distribution of daily rainfall is likely to be.

[1]

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(c) The data show that 35% of days are “wet” (rainfall of at least 1 mm). Using a suitable model, find the probability that at least 4 days in a week of 7 days are wet, and state one assumption you have made.

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(Total for Question 25 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

26. The probability that a component is faulty is 0.15, independently of other components. Find the smallest number of components, n , that must be tested so that the probability of finding at least one faulty component is greater than 0.99. [4]

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END OF WORKSHEET

Total: 103 marks